Major Currencies
FX Market Decision Support System

Data Science Project Report

Prepared by

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Chapter 1

Introduction

This report presents the development of a decision-support system for the Forex market, with a focus on major currency pairs. The project aims to address the challenges of decision-making in a highly volatile and information-rich environment by adopting a multi-agent and data-driven approach. By integrating technical analysis, macroeconomic indicators, market sentiment, event impact assessment, and risk modeling, the proposed system seeks to generate reliable, actionable, and explainable trading signals to support traders and analysts.

Chapter 2

Phase 1: Business Understanding and Analytic Approach

2.1 Objective

The objective of this phase is to clearly define the business problem, identify the project's strategic objectives, and establish measurable success criteria. This phase ensures alignment between business needs and the analytical approach adopted for the project.

2.2 Problem Definition

Decision-making in Forex trading is complex due to market volatility, rapid information flow, and the influence of multiple economic and behavioral factors. Existing solutions typically rely on a single analytical dimension, such as technical indicators or economic news, resulting in fragmented, subjective, and often unreliable trading decisions.

2.3 Project-Specific Problem

The main challenge lies in integrating heterogeneous sources of information—price data, macroeconomic indicators, market sentiment, and event-driven signals—into a unified and explainable decision-support framework. Traders often lack tools that provide both reliable signals and clear justifications for those signals.

2.4 Business Objectives

The business objectives of the project are:

- Reduce false trading signals generated by single-source analytical tools.
- Improve the reliability of Forex decision-making through multi-source integration.
- Provide explainable and actionable trading signals to increase user trust and adoption.

- Automate market analysis while maintaining transparency and interpretability.

2.5 Success Criteria

The success of the project will be evaluated using the following criterias

- Signal accuracy measured through win rate and prediction performance.
- Reduction of false signals via cross-validation between analytical agents.
- Clarity and consistency of signal explanations.
- Positive adoption and perceived usefulness by traders.

2.6 Analytic Approach

To address the identified business problem, a structured analytic approach is adopted. The system is designed around a **multi-agent architecture**, where each agent focuses on a specific analytical dimension:

- Technical analysis to capture trends and price patterns.
- Macroeconomic analysis to assess the impact of economic indicators and monetary policy.
- Market sentiment analysis to capture investor psychology from news and positioning data.

Each agent produces a signal with an associated confidence score. Signals are validated through a **cross-validation mechanism (Four-Eyes Principle)** to reduce noise and inconsistencies. Additionally, the system supports **event-driven analysis**, allowing users to evaluate the impact of specific events on currency movements. **Monte Carlo simulations** are used to model uncertainty and assess potential risk scenarios.

2.7 Project Scope

2.7.1 In Scope

The project focuses on major currency pairs in the Forex market and provides analytical decision support based on multi-source data integration. The system is designed as an advisory tool and does not perform automatic trade execution.

2.7.2 Out of Scope

Real-money trading execution, broker integration, regulatory compliance, and high-frequency trading strategies are outside the scope of this project.

2.8 Stakeholders and Target Users

- Retail traders seeking reliable and explainable decision support.
- Professional traders and financial analysts.
- Data science researchers and academic supervisors.

2.9 Assumptions and Constraints

2.9.1 Assumptions

- Historical market and macroeconomic data are reliable and accessible.
- Market patterns can be partially captured through data-driven models.
- Users rely on the system as a decision-support tool rather than an autonomous trader.

2.9.2 Constraints

- Limited project timeline and computational resources.
- Dependency on external data sources and APIs.
- Market unpredictability and sudden geopolitical events.

2.10 Risks and Mitigation

- **Risk:** Model overfitting. **Mitigation:** Cross-validation and backtesting.
- **Risk:** Noisy or incomplete data. **Mitigation:** Data quality monitoring and filtering.
- **Risk:** Misinterpretation of signals. **Mitigation:** Clear explanations and confidence indicators.

2.11 Conclusion of Phase 1

This phase established a clear understanding of the business problem and defined a robust analytic approach aligned with business objectives. The outcomes of this phase provide a solid foundation for the next steps of the project, which will focus on data understanding and exploration.

Chapter 3

Phase 2: Data Acquisition and Data Understanding

3.1 Data Acquisition

This phase focuses on collecting and structuring heterogeneous datasets required for building the Forex decision-support system. Data was gathered from multiple reliable sources to ensure robustness and analytical depth.

3.1.1 Forex Market Data

Historical price data was collected for the following major currency pairs:

- EUR/USD
- USD/JPY
- GBP/USD
- USD/CHF

The dataset includes OHLC prices, trading volume, and timestamped records (UTC). Data was retrieved using the MetaTrader 5 (MT5) API and structured for multi-timeframe analysis (1H and 1D).

3.1.2 Macroeconomic Data

Macroeconomic indicators were retrieved from the FRED (Federal Reserve Economic Data) API. The selected indicators include:

- Consumer Price Index (CPI)
- Federal Funds Rate
- Unemployment Rate

These indicators were aligned temporally with FX price data to study macro-price relationships.

3.1.3 News and Sentiment Data

Financial news data was collected and processed using Natural Language Processing techniques. Sentiment scores were generated to quantify market psychology and integrated as numerical features in the modeling phase.

3.2 Data Understanding and Exploratory Analysis

Extensive exploratory data analysis (EDA) was conducted to understand statistical properties, correlations, volatility dynamics, and predictive structure of the data.

3.2.1 Currency Pair Correlation Analysis

The correlation matrix of currency pair returns is shown in Figure 3.1. Strong relationships were observed, particularly a dominant USD factor structure.

Figure 3.1: Correlation Matrix of Major Currency Pair Returns

Notably:

- EUR/USD and GBP/USD show strong positive correlation.
- EUR/USD and USD/CHF exhibit strong negative correlation.

- USD-driven structural relationships are evident.

These findings highlight the importance of portfolio-level risk management and factor-based modeling.

3.2.2 Volatility Comparison Across Currency Pairs

Figure 3.2 compares average normalized volatility across currency pairs.

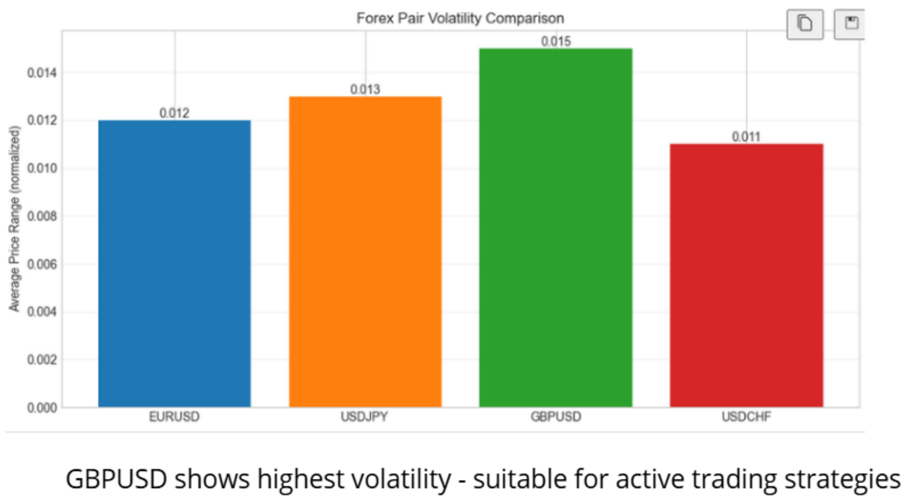

Figure 3.2: Forex Pair Volatility Comparison

GBP/USD exhibits the highest volatility, suggesting suitability for active trading strategies, while USD/CHF shows relatively lower volatility.

3.2.3 Macroeconomic Indicators Correlation

The correlation matrix between macroeconomic indicators is presented in Figure 3.3.

A strong inverse relationship between CPI and Federal Funds Rate was observed, confirming macroeconomic interdependencies that may influence currency dynamics.

3.2.4 Feature Importance Analysis

Feature importance results from preliminary modeling are presented in Figure 3.4.

Key predictive contributors include:

- Sentiment scores
- MACD
- RSI
- ATR

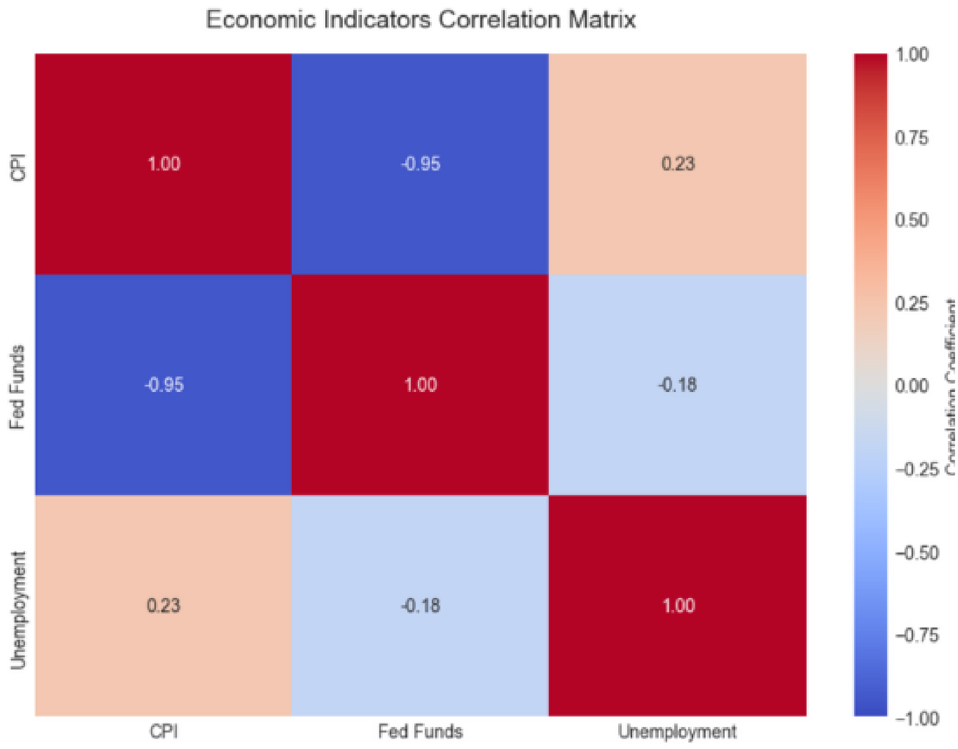

Figure 3.3: Macroeconomic Indicators Correlation Matrix

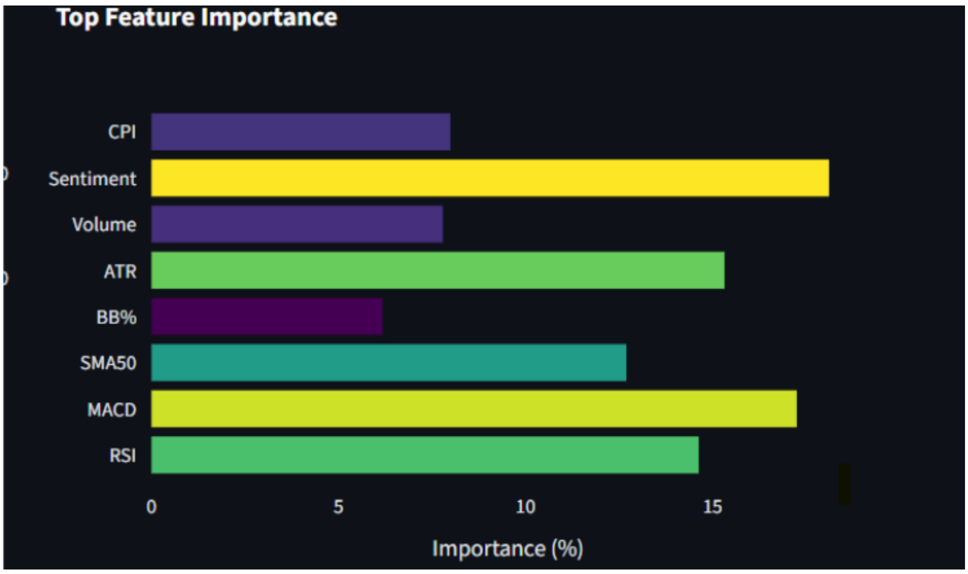

Figure 3.4: Top Feature Importance Ranking

- CPI

This confirms the multi-dimensional nature of predictive modeling in Forex markets.

3.2.5 Data Readiness Assessment

A Data Science Optimization (DSO) readiness assessment was conducted to evaluate infrastructure and data preparation maturity.

Figure 3.5: DSO Readiness Assessment

The system demonstrates high readiness in Forex data acquisition and macroeconomic integration, with ongoing improvements in agent design and dashboard components.

3.2.6 Data Quality Evaluation

Data quality metrics were evaluated across three categories: Forex data, Economic data, and News data (Figure 3.6).

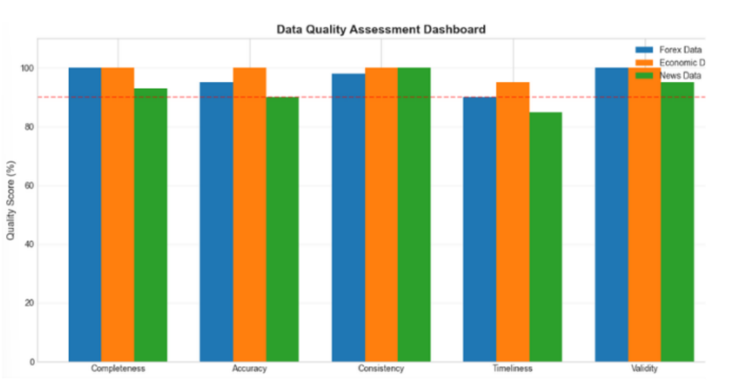

Figure 3.6: Data Quality Assessment Dashboard

Evaluation criteria included:

- Completeness
- Accuracy

- Consistency
- Timeliness
- Validity

Overall quality scores exceeded 85%, indicating strong data reliability for modeling purposes.

3.3 Key Insights from Phase 2

The data exploration phase revealed several critical findings:

- FX returns exhibit weak linear autocorrelation, supporting the need for non-linear machine learning models.
- Strong cross-pair correlations indicate USD factor dominance.
- Volatility clustering confirms the relevance of volatility-aware modeling.
- Returns display heavy-tailed distributions, invalidating Gaussian assumptions.
- Multi-source features (technical, macro, sentiment) significantly enhance predictive capability.

3.4 Conclusion of Phase 2

Phase 2 established a comprehensive understanding of the statistical and structural properties of Forex market data. The insights gained justify the adoption of a multi-agent, volatility-aware, and machine learning-based framework for signal generation. These findings provide a strong foundation for Phase 3, which will focus on modeling strategy and system implementation.